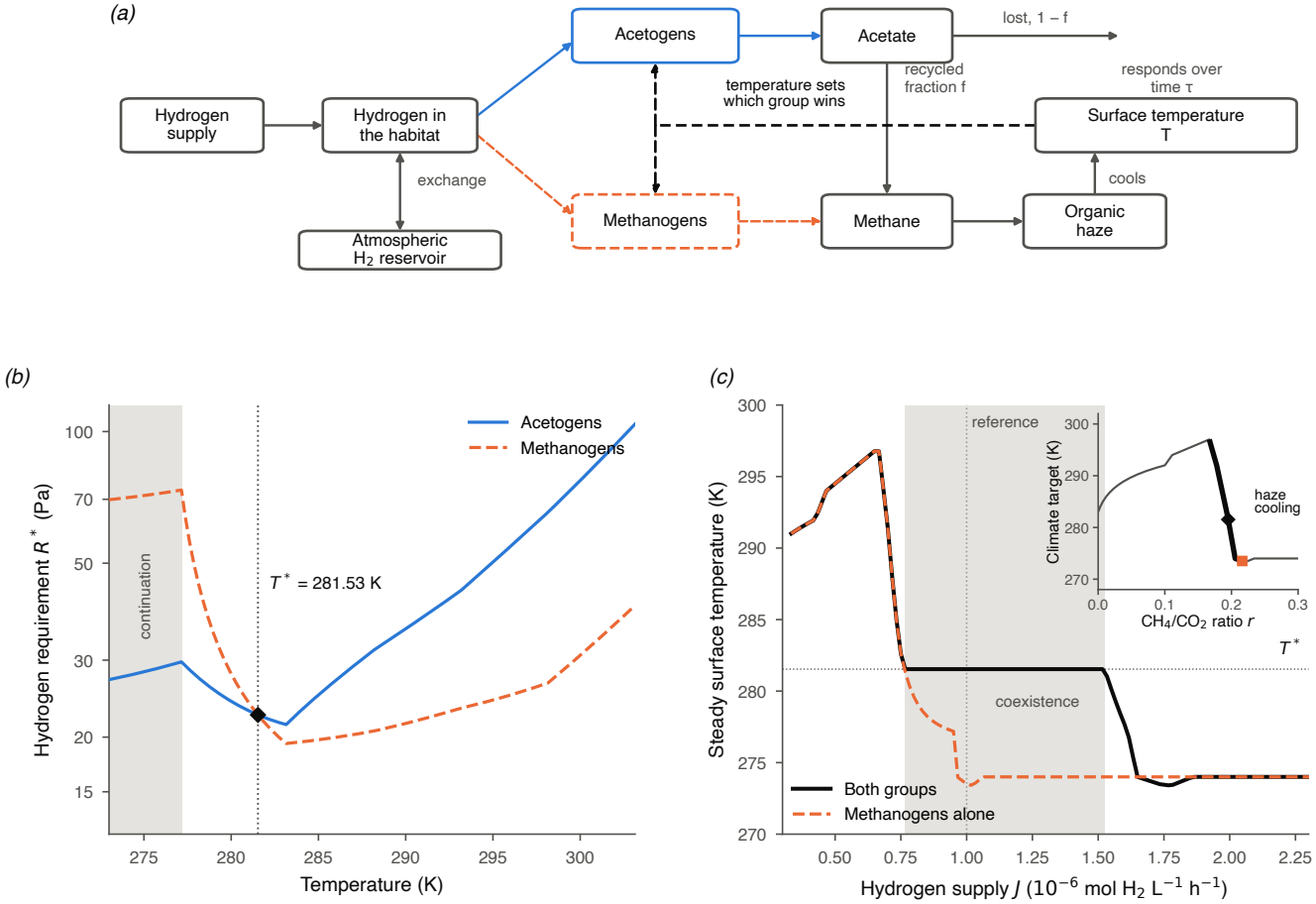
(a)
Acetogens
Acetate
lost, 1 – f
recycled fraction f
responds over time τ
temperature sets which group wins
Hydrogen supply
Hydrogen in the habitat
Surface temperature T
exchange
cools
Methanogens
Methane
Organic haze
Atmospheric H_2 reservoir
(b)
Acetogens
Methanogens
Hydrogen requirement R^* (Pa)
continuation
T^* = 281.53 K
Temperature (K)
(c)
reference
Climate target (K)
haze cooling
CH_4/CO_2 ratio r
T^*
Steady surface temperature (K)
coexistence
Both groups
Methanogens alone
Hydrogen supply J (10^{-6} mol H_2 L^{-1} h^{-1})